

BURLINGTON PIERS, ON, Canada

WMO: 714370

Lat: **43.3005N**Lon: **79.7916W**Elev: **77**StdP: **100.40**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.9	-13.3	-21.6	0.5	-14.9	-19.2	0.7	-12.4	12.4	-4.1	11.3	-3.8	4.4	0	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	30.3	21.4	28.7	20.7	27.1	20.1	23.2	27.4	22.3	26.4	21.6	25.5	4.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.0	16.8	25.6	21.0	15.8	24.5	20.2	15.0	24.0	69.5	27.3	66.0	26.5	63.2	25.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.2	8.0	-18.4	32.7	3.2	1.5	-20.8	33.7	-22.7	34.6	-24.5	35.5	-26.8	36.6		
(5)			-19.0	24.8	3.0	1.5	-21.2	25.8	-22.9	26.7	-24.6	27.5	-26.9	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-3.2	-2.6	1.3	7.0	12.9	18.8	21.9	21.6	18.1	11.6	5.6	0.4
	DBStd	9.90	5.83	5.22	4.92	3.99	4.24	3.94	3.07	2.72	3.57	4.05	4.34	4.67
	HDD10.0	1627	411	354	271	108	14	0	0	0	0	28	143	298
	HDD18.3	3547	668	587	527	341	177	42	4	4	46	212	382	556
	CDD10.0	1451	0	0	3	17	106	263	371	358	244	78	10	1
	CDD18.3	329	0	0	0	0	11	56	116	104	40	3	0	0
	CDH23.3	1748	0	0	0	2	74	326	695	474	165	12	0	0
	CDH26.7	419	0	0	0	0	12	87	188	97	33	1	0	0
Wind	WSAvg	3.7	4.3	4.1	4.2	4.0	3.3	3.1	3.0	3.1	3.5	3.9	4.0	4.2
Precipitation	PrecAvg	884	71	62	64	76	80	79	82	77	77	73	73	69
	PrecMax	1102	148	115	140	145	146	196	187	159	173	150	133	137
	PrecMin	686	30	33	27	31	26	23	18	30	25	17	21	36
	PrecStd	109	29	22	31	30	30	40	42	31	34	33	28	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	12.7	18.0	22.9	28.3	31.3	32.6	31.6	30.0	24.9	18.3	13.7
		MCWB	10.6	9.3	12.1	14.5	18.6	21.4	22.7	21.9	21.8	17.8	13.6	11.1
	2%	DB	9.0	8.7	13.7	19.1	25.5	29.3	30.5	29.1	27.2	22.0	16.0	10.8
		MCWB	6.9	5.6	9.3	12.7	17.5	20.7	21.5	21.5	20.2	16.5	12.1	8.5
	5%	DB	6.2	5.5	10.5	15.8	22.7	27.0	28.8	27.5	25.1	19.7	14.0	8.0
		MCWB	4.2	2.9	7.1	10.4	16.2	19.7	20.7	20.6	19.4	15.4	10.4	5.7
	10%	DB	3.9	3.6	7.2	12.8	19.9	24.9	27.1	26.0	23.4	17.5	11.7	6.0
		MCWB	2.4	1.7	4.5	8.2	14.5	18.8	20.2	20.1	19.1	14.1	8.8	4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	10.2	13.7	15.7	20.7	23.5	25.0	24.2	23.1	20.1	14.9	11.4
		MCDB	12.6	12.1	17.5	20.1	26.1	28.2	29.5	28.0	27.0	23.3	17.3	13.3
	2%	WB	7.3	6.0	10.2	13.6	18.8	22.0	23.5	22.9	21.8	17.5	12.9	8.7
		MCDB	8.6	8.5	12.7	18.2	23.7	26.7	27.8	26.8	25.2	20.4	15.5	10.5
	5%	WB	4.2	3.4	7.2	10.9	17.0	20.9	22.5	22.0	20.8	15.9	10.7	6.1
		MCDB	5.6	5.1	10.1	15.4	21.6	25.3	26.6	25.8	24.0	18.6	13.4	7.6
	10%	WB	2.5	1.9	4.6	8.7	15.1	19.7	21.5	21.3	19.7	14.6	9.0	4.3
		MCDB	3.8	3.3	6.8	12.5	19.2	24.1	25.6	24.9	22.6	17.4	11.4	5.8
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.3	6.2	6.8	7.4	7.0	6.9	6.2	6.5	6.1	5.4	5.0
		MCDBR	8.1	8.6	10.4	12.8	12.0	10.2	9.2	8.4	8.9	9.5	8.9	7.3
		MCWBR	7.5	7.3	7.7	8.1	6.5	5.0	4.3	4.2	4.7	6.0	6.9	6.4
	5% WB	MCDBR	7.7	8.1	9.9	12.0	11.6	9.1	8.0	6.9	7.1	8.2	8.3	7.3
MCWBR		7.5	7.3	8.0	8.7	7.5	5.3	4.6	4.4	4.7	6.1	6.9	6.8	
Clear-Sky Solar Irradiance	taub	0.296	0.335	0.338	0.386	0.409	0.446	0.437	0.422	0.389	0.366	0.315	0.299	
	taud	2.403	2.264	2.388	2.289	2.278	2.201	2.249	2.280	2.353	2.396	2.507	2.450	
	Ebn at Noon	838	851	903	879	865	831	833	834	836	809	814	801	
	Edh at Noon	77	104	105	126	131	142	134	126	108	92	70	68	
All-Sky Solar Radiation	RadAvg	1.41	2.16	3.40	4.43	5.39	5.90	5.90	5.19	4.15	2.51	1.57	1.09	
	RadStd	0.14	0.21	0.36	0.48	0.56	0.42	0.43	0.27	0.39	0.23	0.21	0.12	